

Hi everyone,

This is the solution to today's problem. I have added both single-line and multi-line comments to help you understand the code better and improve readability.

I hope you enjoyed the session and learned something new.

I have also given you comment access, so if you have any questions about any line of code, feel free to ask here.

Thank you.

```
"""
```

```
Program: Hospital Karel
```

```
Karel traverses 1st Street from west to east, building hospitals  
wherever it encounters a beeper.
```

```
"""
```

```
from karel.stanfordkarel import *
```

```
def main():
```

```
    while front_is_clear():  
        if beepers_present():  
            build_hospital()  
        safe_move()
```

```
"""
```

```
    Karel picks up supplies and builds a hospital.  
    Pre-condition: Karel is on a beeper, representing a  
        pile of supplies. Karel is facing east.  
    Post-condition: Karel is standing at the bottom  
        of the last column of the hospital, facing east.
```

```
"""
```

```
    # pick up supplies
```

```
def build_hospital():
```

```
    pick_beeper()  
    do_one_column()  
    move()
```

```

do_one_column()

"""
Karel builds a single column of a hospital.
Pre-condition: Karel is facing east at the bottom
                of where we want to build a column.
Post-condition: Karel is facing east at the bottom
                of the column it just built.
"""

def do_one_column():
    turn_left()
    put_three_beeper()
    return_to_base()
    turn_left()

"""
Karel places three beepers in a row.
Pre-condition: Karel is on the corner where we want
                to place the first beeper.
Post-condition: Karel is on the corner where it
                placed the third beeper in a row.
"""

def put_three_beeper():
    put_beeper()
    move()
    put_beeper()
    move()
    put_beeper()

"""
Karel turns around and goes to the wall.
Pre-condition: Karel is at the end of the column
                it just built, facing north.
Post-condition: Karel has returned to 1st Street,
                below the column is just built, facing south.
"""

def return_to_base():
    turn_around()
    move_to_wall()

```

```
def move_to_wall():  
    while front_is_clear():  
        move()
```

```
def safe_move():  
    if front_is_clear():  
        move()
```

```
def turn_around():  
    turn_left()  
    turn_left()
```

```
if __name__ == '__main__':  
    main()
```